

ECE298A

Integrated Circuit Design and Tapeout

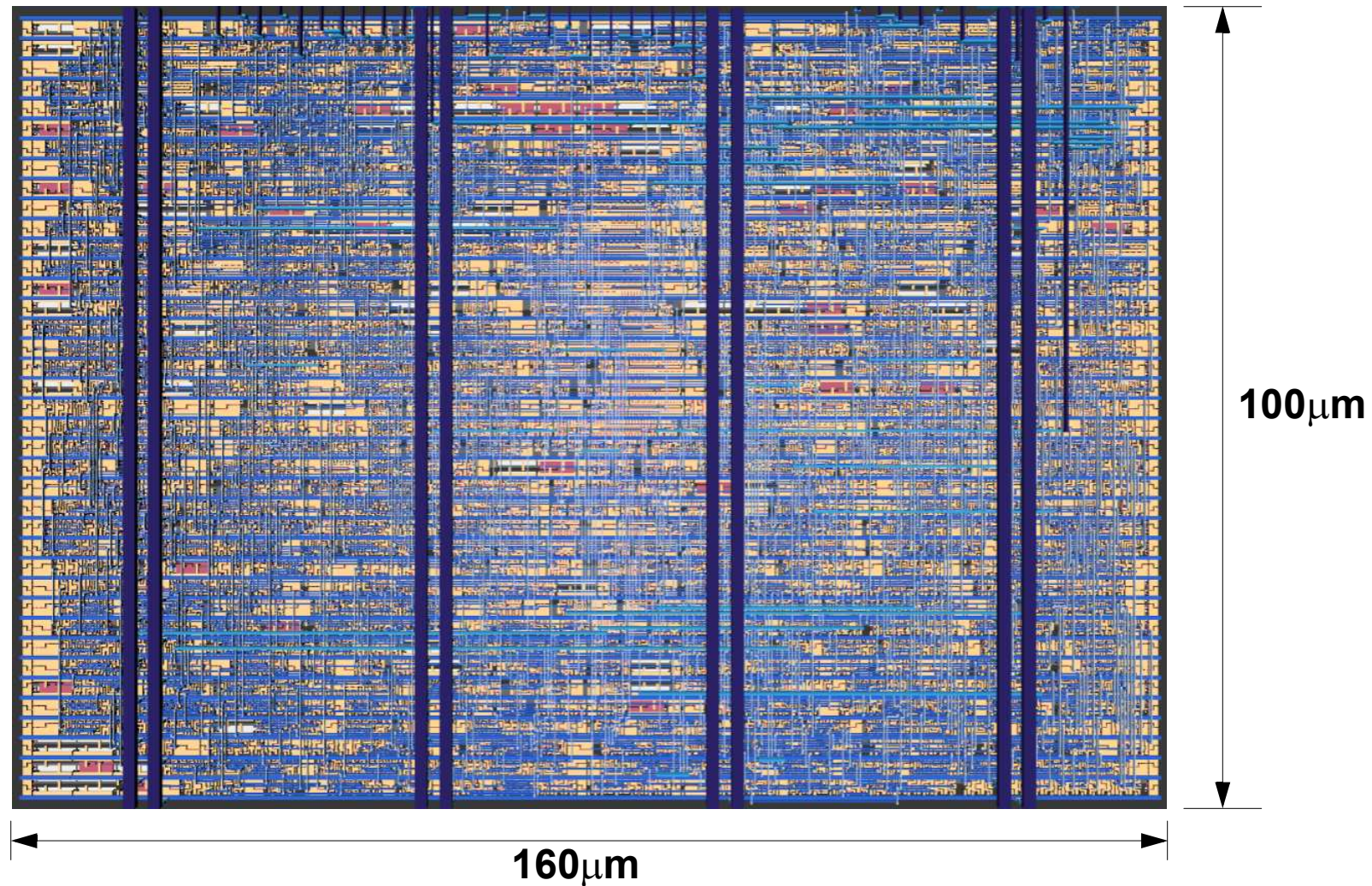
John R. Long
RF/MMIC Group
University of Waterloo
jrlong@uwaterloo.ca

September 10, 2026

Introduction

- ECE 298A is a project course for self-driven students, where you will build your own silicon chip using open-source tools and 130-nm or 180-nm CMOS design kits.
- ECE 298A follows a real-world chip design flow from hardware description to fabrication. Learning goals are to become familiar with the design flow for making an IC:
 1. Concept development
 2. Analysis, design and simulation
 3. Physical layout and verification
- Potential projects: bit-serial processor, TT-RAM, + ?
- Designs are sent for fabrication at the end of the term (Sky26d or GF26c tapeout). Delivery expected in late 2027 (7 months fabrication time for Sky130, 3 months for GF + 2 months for testboard fabrication, assembly and delivery).

Chip from F2024



- 8-bit RISC CPU: <https://github.com/gjrchen/8-Bit-CPU-top>

Open-Source Design Flow

- No restrictive licenses or non-disclosure agreements (NDAs) required.
- Tools run in the cloud. Software is not installed locally on your computer (although you can do that too, e.g., icarus verilog).
- Each chip contains a copy of every design submitted, so any design can be selected and tested.
- Everyone shares the finished chip and circuit board. Chips are mounted on a demo board for ease of testing using a μ -python interface.
- Cost of fabrication is kept low by sharing the chip space with ~200 other designs. Fabrication and prototyping board costs are paid by the ECE department.
- IP is easily shared on github.

Project Timeline

Week	Topic
1 (Sept 10)	Introduction; github registration and github actions
2 (Sept 17)	Teams formed and projects defined; HDL and standard-cell design and synthesis
3 (Sept 24)	Design in Verilog. Task 1: Design proposal
4 (Oct 1)	Design in Verilog; test using CocoTB
5 (Oct 8)	Task 2: Verilog design and test
-	Reading week (October 12-15)
6 (Oct 22)	Design in Verilog; test using CocoTB
7 (no meet-up)	Midterm week (October 26-30)
8 (Oct 22)	Implementation and timing verification
9 (Nov 5)	Implementation and timing verification Task 3: Verilog evaluation
10 (Nov 12)	Synthesis, parasitic extraction and back annotation
11 (Nov 19)	Synthesis, parasitic extraction, and design verification with Open Road
12 (Nov 26)	Final verification (github actions complete and Tiny Tapeout submission ready) Task 4: System verification
13 (Dec 3)	Final submissions and documentation complete

Team Formation and Project Definition

- You work in teams of two. Each team proposes a project and submits a design for fabrication. Teams should be formed by September 17th (our next meet-up).
- Potential projects:
 - RISC-V
 - bit-serial processor
 - TT-RAM (gen-2)
 - digital radio
 - DSP: filter, equalizer, etc.
- Outline your project topic by September 17th; develop a preliminary specification by Sept 24th.
- TT provides: 8 inputs, 8 outputs, 8 bidirectional I/Os, clock, reset, power and ground. Digital blocks are coded in verilog and tested using CocoTB.
- On September 24th we'll assess your group's project plans and preliminary spec.

Other 'To Do' Things

- Software skills: Verilog and Python.
- Communicate using Slack or Discord. Join the TT Discord channel
- First task: an 8-bit programmable counter with synchronous load and tri-state outputs. Build using github actions in Tiny Tapeout for the GF-180nm technology.

Tiny Tapeout Github Actions

1. Create a user account on github and sign in
2. From the 'overview' page:
 - Select 'import repository' (upper right, under '+' pulldown)
 - <https://github.com/TinyTapeout/ttgf-verilog-template>
 - Make the repository '**Public**' so you can run github actions
3. Review '**src/project.v**' and edit '**info.md**' under **docs** directory
4. Edit 'info.yaml' : header with 'project title,' 'author,' and include I/O pin names). 'Commit changes' to save the file.
- 5.) Enable github actions:
 - Settings --> 'Pages' (under 'Build and Deployment, Source' - change from '**Deploy from a branch**' to '**GitHub Actions**')
 - docs and test jobs fail (e.g., if docs or test are not updated)
 - gds job succeeds!
6. Click through 'Update info.yaml' to view results (takes up to 15 min. to run)
 - 'View job summary' under 'viewer summary' and look at 3D view
7. Check other output files (e.g., test, TT submission zipfile, etc.)